card usage. Also, they use clustering to identify clusters of customers – for instance, to find loyal customers versus customers who have stopped using their services.

In medicine, clustering can be used to characterise patient behaviour, based on similar characteristics, so as to identify successful medical therapies for different illnesses. In biology, clustering is used to group genes with similar expression patterns or to cluster genetic markers to identify family ties. If you look around, you can find many other applications of clustering, but generally, it can be used for one of the following purposes: exploratory data analysis, summary generation or reducing the scale, outlier detection (especially to be used for fraud detection or noise removal), finding duplicates and data sets; or as a pre-processing step for either prediction, other data mining tasks or as part of a complex system.

## A brief look at different clustering algorithms and their characteristics

Partitioned based clustering is a group of clustering algorithms that produces clusters, such as k-means, k-medians or fuzzy c-means. These algorithms are quite efficient and are used for medium and large sized databases. Hierarchical clustering algorithms produce trees of clusters, such as agglomerative and divisive algorithms. These groups of algorithms are very intuitive and are generally good with small-sized data sets. Density based clustering algorithms produce arbitrary shaped clusters. They are especially good when dealing with spatial clusters or when there is noise in your data set — for example, the DBSCAN algorithm.

In this article, I will take you through the types of clustering, the different clustering algorithms and compare two of the most commonly used clustering methods.

## k-means clustering

k-means is probably the most well-known clustering algorithm. It divides data into non-overlapping subsets (clusters) without any cluster internal structure. The objective of k-means is to form clusters in such a way that similar samples go into a cluster and dissimilar samples fall into different clusters. So we can say k-means tries to minimise the intra-cluster distances and maximise the inter-cluster distances.

Now the question is about how to calculate the dissimilarity or distance of two cases such as two customers. We can easily use a specific type of Minkowski distance to calculate the distance of these two customers. Indeed, it is the Euclidean distance.

And what if we have more than one feature? Well, we can still use the same formula, but this time, in a two-dimensional space.

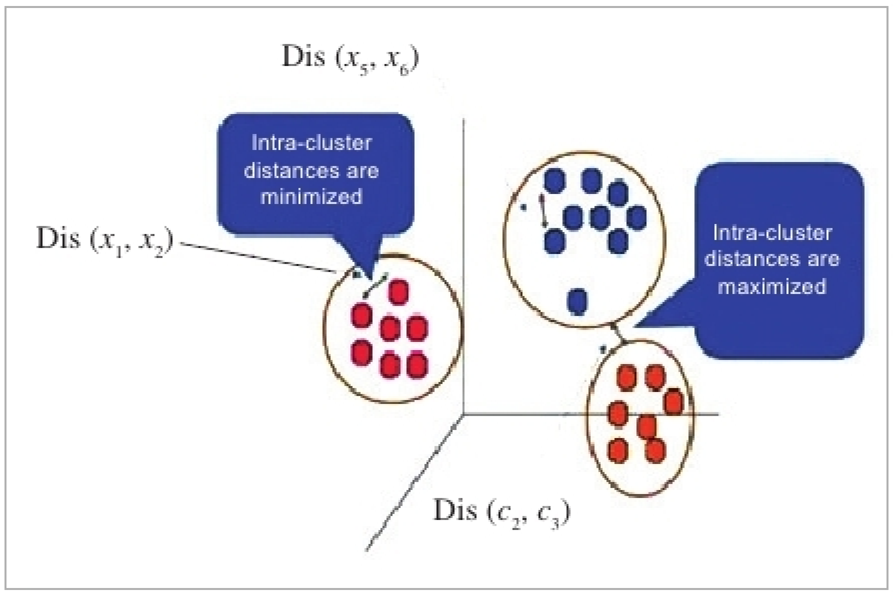


Figure 1: k-means clustering

$$\text{Dis}\ (x_1, x_2) = \sqrt{\sum_{i=0}^{n}(x_{1i} - x_{2i})^2}$$

Figure 2: Euclidean distance

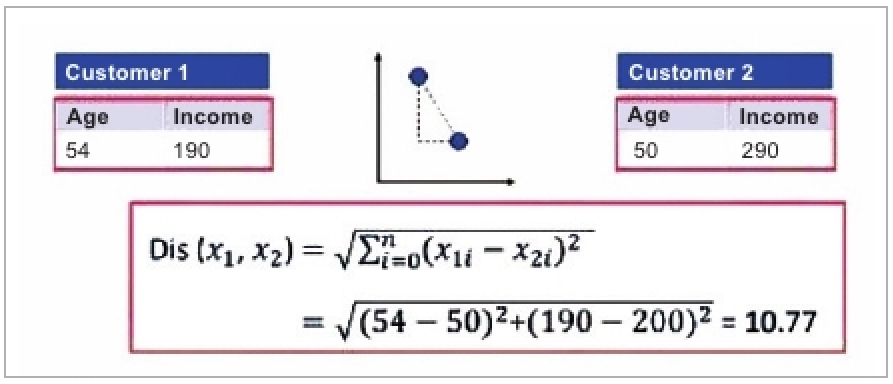


Figure 3: Euclidean distance in 2D

The steps involved are listed below.

1. In the first step, we should determine the number of clusters. The key concept of the k-means algorithm is that it randomly picks a centre point for each cluster, which means that we must initialise 'k', which represents the number of clusters.
2. Then, the algorithm will randomly select the centroids of each cluster.
3. After the initialisation step, which defined the centroid of each cluster, we have to assign each point to the closest centre. For this purpose, we have to calculate the distance of each data point or in our case, each customer, from the centroid points, which will be assigned for each data point to the closest centroid (using Euclidean distance).
   It is safe to say that this does not result in good clusters because the centroids were chosen randomly from the beginning. Indeed, the model will probably have a high error rate. Here, the error is the total distance of each point from its centroid.
4. So how can we turn them into better clusters with less error? We move the centroids. The main objective of